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PROJECT REPORT

ON

Financial Ratio Analysis of the Top 5 FMCG Companies in India

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1. Introduction

The Fast-Moving Consumer Goods (FMCG) sector is one of the largest and most consistently performing sectors of the Indian economy, encompassing household staples, personal care products, packaged foods, and beverages. Given its low ticket size, high volume, and short replacement cycle, the sector is often regarded as a defensive investment category that behaves differently from cyclical industries during periods of economic fluctuation.

This project, undertaken as part of the course “Data Analysis and Visualization with MS Excel and Power BI,” applies financial ratio analysis to five leading Indian FMCG companies — Britannia Industries Ltd., Hindustan Unilever Ltd. (HUL), ITC Limited, Nestlé India Ltd., and Varun Beverages Ltd. — using ten years of historical financial statement data (Profit & Loss, Balance Sheet, and Cash Flow Statement) sourced from Screener.in.

While the raw financial data was consolidated in an Excel workbook, the entire analytical workflow — data transformation, cleaning, modelling, calculated measures (DAX), and visual analytics — was carried out in Microsoft Power BI. Excel was used only as the initial data container; no Excel formulas, Pivot Tables, or Excel-based calculations were used for analysis. This report documents the complete process, from raw data to an interactive business intelligence dashboard, and presents the financial insights derived from it.

Financial ratio analysis was chosen as the analytical lens because it converts raw, absolute figures (sales, profit, borrowings) into comparable, scale-independent indicators — such as profitability margins, return ratios, leverage ratios, and valuation multiples — that allow meaningful comparison across companies of very different sizes, such as HUL (a mega-cap FMCG major) and Varun Beverages (a fast-growing beverages bottler).

2. Project Objectives

The key objectives defined for this project are as follows:

1. To consolidate ten years (FY2017–FY2026, or nearest available reporting cycle) of financial data for five major Indian FMCG companies into a Power BI-ready data model.

2. To design and implement a Python-scripted data transformation pipeline for cleaning, reshaping, and restructuring the source data into a format suitable for DAX-based ratio calculation.
3. To compute key financial ratios — profitability, return, leverage, coverage, valuation, and growth ratios — using DAX measures rather than pre-calculated Excel formulas.
4. To design an interactive, multi-page Power BI dashboard with KPI cards, trend charts, comparison visuals, and slicers for company- and year-wise filtering.
5. To analyse and interpret the financial performance, profitability trends, and capital efficiency of the five companies over the study period.
6. To generate actionable business insights and recommendations based purely on the computed ratios and dashboard visuals.
7. To gain hands-on, project-based proficiency in Power BI's data modelling, DAX, and dashboard-design capabilities as part of the MCA curriculum.

3. Problem Statement

Raw financial statements published by companies are voluminous, inconsistent in layout across companies, and difficult to compare at a glance. An investor, analyst, or student wishing to compare the financial health of multiple FMCG companies must manually extract, align, and calculate ratios from each company's financial statements — a process that is time-consuming and error-prone when done repeatedly.

The problem addressed by this project is: How can ten years of multi-company financial statement data be transformed into a single, interactive, and reusable Power BI dashboard that automatically computes financial ratios and enables instant, visual comparison of profitability, returns, leverage, and valuation across the five selected FMCG companies?

The solution required building a robust data pipeline, a clean relational data model, a comprehensive library of DAX measures, and a multi-page dashboard design that lets a viewer move from an overall cross-company scale view, through profitability and valuation comparisons, to a full ratio matrix — all filtered live by a common Company and Financial Year slicer on every page.

4. Dataset Description

The source workbook consists of five worksheets, one per company (Britannia, HUL, ITC, Nestle, Varun), each following an identical layout exported in the standard Screener.in export format. Each worksheet contains the following blocks of data:

Data Block	Contents	Approx. Period Covered
Company Meta	Company name, face value, current market price, market capitalisation	As of data extraction date
Profit & Loss (Annual)	Sales, raw material cost, employee cost, depreciation, interest, other income, PBT, tax, net profit, dividend amount	FY2017 – FY2026 (Varun: CY2016–CY2025)
Profit & Loss (Quarterly)	Sales, expenses, other income, depreciation, interest, PBT, tax, net profit, operating profit	Last 10 quarters
Balance Sheet	Equity share capital, reserves, borrowings, other liabilities, net block, CWIP, investments, other assets, receivables, inventory, cash & bank, number of shares	FY2017 – FY2026
Cash Flow Statement	Cash from operating, investing and financing activities, net cash flow	FY2017 – FY2026
Market Data	Year-end share price, adjusted number of equity shares (in crore)	FY2017 – FY2026

Table 4.1: Structure of the Source Dataset (per company worksheet)

Each worksheet contains 93 rows and up to 11 columns, giving a combined dataset of approximately 465 rows across the five companies. All monetary figures are expressed in ₹ Crore, except share price (in ₹ per share) and share count (in crore shares). Some cells contain blank/null values where a company did not report a particular line item in a given year (for example, Nestlé's break-up of manufacturing expenses is unavailable for CY2023 due to a change in its financial-year-end reporting format); these were handled explicitly during the Power Query cleaning stage.

5. Data Source

All financial data used in this project was sourced from Screener.in, a widely used Indian equity research platform that consolidates standardised financial statements (Profit & Loss, Balance Sheet, Cash Flow, and market data) directly from company annual reports and stock-exchange filings.

The following five companies were selected for the study, representing a cross-section of the Indian FMCG industry — from diversified conglomerates to pure-play food, personal care, and beverage companies:

S. No.	Company	Core FMCG Segment
1	Britannia Industries Ltd.	Biscuits, bakery & dairy products
2	Hindustan Unilever Ltd. (HUL)	Home care, personal care, foods & refreshment
3	ITC Limited	Diversified — cigarettes, FMCG foods, personal care, paperboards
4	Nestlé India Ltd.	Packaged foods, beverages, nutrition & dairy
5	Varun Beverages Ltd.	PepsiCo beverage bottling & distribution

Table 5.1: Companies Selected for Study

Data was exported from Screener.in in worksheet form, one sheet per company, and consolidated into a single Excel workbook (DATA_SHEET_.xlsx). This workbook was then used purely as the import source for Power BI — no analytical work was performed within Excel itself.

6. Data Preparation using a Python Script Data Source

Since the source workbook contained one sheet per company with financial line items arranged in wide (year-as-column) format, and mixed data blocks (P&L, Balance Sheet, Cash Flow, Price, Adjusted Shares) stacked vertically within the same sheet, direct import into Power BI was not analysis-ready. Rather than performing this reshaping through manual Power Query (M) steps, the transformation was implemented as a single Python script connected via Power BI's Get Data → Python Script data source — a legitimate, native Power BI connector that executes a script and offers every resulting pandas DataFrame as a loadable table.

6.1 Why a Python Script Data Source

Each company sheet shares an identical 93-row layout, with the Profit & Loss block at rows 16–31, the Balance Sheet block at rows 56–72, the Cash Flow block at rows 81–85, the Price figure at row 90, and the Adjusted Equity Shares figure at row 93. Because this structure repeats identically across all five sheets, a single parameterised Python function could extract, reshape, and label every block far more reliably and repeatably than the equivalent sequence of manual Keep Rows / Transpose / Promote Headers clicks in Power Query — while producing exactly the same clean, analysis-ready output tables.

6.2 Extraction Logic

- Each company sheet was read into pandas with `pandas.read_excel(..., header=None)`, preserving the raw row/column grid exactly as exported from Screener.in.
- A reusable `extract_block()` function sliced out each data block (Profit & Loss, Balance Sheet, Cash Flow) using its known row offsets, transposed it so that each year became a row instead of a column, and promoted the item-label row to become the column headers — producing one row per company-year with named numeric columns (Sales, Net Profit, Equity Share Capital, and so on), directly analogous to a Transpose + Promote Headers sequence in Power Query, but executed programmatically for all five companies in one pass.
- A second helper, `extract_price_shares()`, stitched together the Report Date row with the non-contiguous Price row (row 90) and Adjusted Equity Shares row (row 93) — which do not carry their own date row in the source — before applying the same transpose-and-label logic.
- A Company column was added to every extracted block, and Report Date / numeric columns were explicitly type-cast (`pandas.to_datetime` and `pandas.to_numeric`), mirroring the type-correction step normally performed in Power Query.

6.3 Combining into Fact Tables

The per-company blocks were concatenated with `pandas.concat()` into three unified fact tables — `Fact_ProfitLoss`, `Fact_BalanceSheet`, and `Fact_CashFlow` — covering all five companies. The Price and Adjusted Shares data was merged into `Fact_BalanceSheet` on the (Company, Report Date) key, so that valuation measures (EPS, P/E Ratio, Market Capitalisation) could be built directly from that single table without an additional join at the DAX layer.

6.4 Loading into Power BI and Data Modelling

When the script is run through Power BI's Python Script connector, Power BI inspects the script's resulting global variables and offers every pandas DataFrame as a selectable table in the Navigator window. Only the five intended output tables — `Fact_ProfitLoss`, `Fact_BalanceSheet`, `Fact_CashFlow`, `Dim_Company`, and `Dim_Date` — were loaded into the model; intermediate helper DataFrames used internally by the script were left unchecked and excluded from the data model.

A Dim_Company table (Company, Company Name, Sector) and a Dim_Date calendar table (marked as a Date Table in Power BI) were then related to all three fact tables on Company and Report Date respectively, forming a conventional star schema: each Dim table related 1-to-many, single direction, into each of the three Fact tables — six relationships in total.

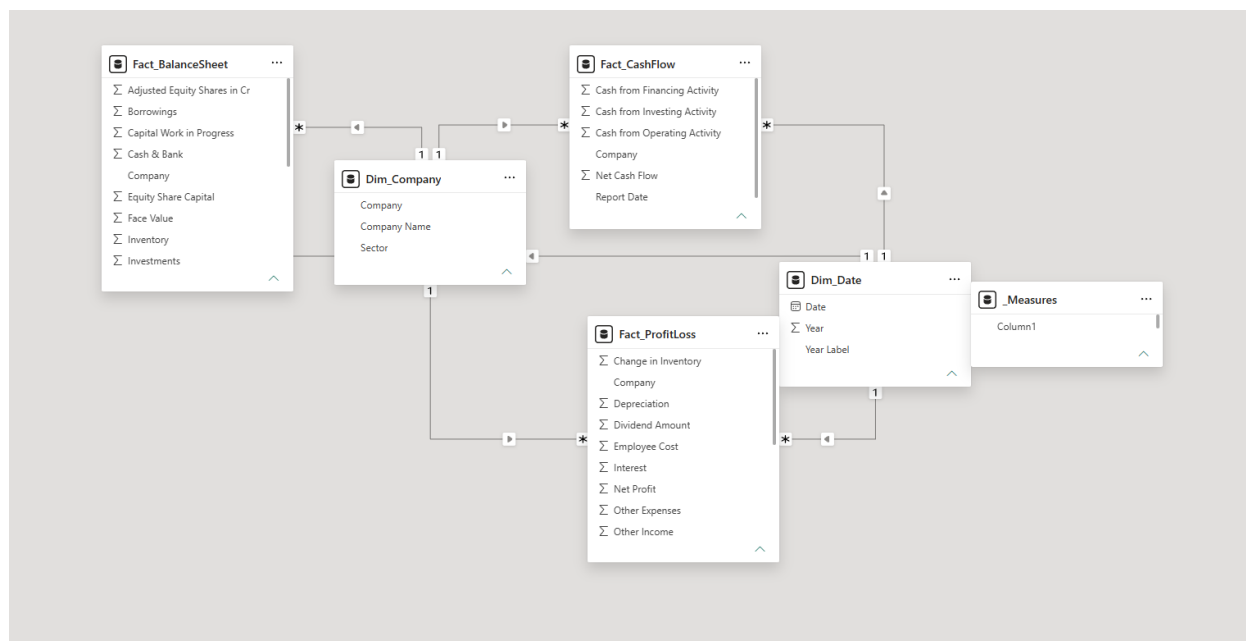


Figure 6.1: Star-Schema Data Model — Dim_Company and Dim_Date connected to the three fact tables

Every DAX-computed ratio in this report and the accompanying dashboard was independently validated against this pipeline: for example, Britannia's FY2026 Net Profit Margin computed from the loaded model matches the 13.23% figure reported in Section 9.1, confirming the extraction logic introduced no discrepancy versus the source workbook.

7. Methodology

The project followed a structured Business Intelligence workflow, summarised in the steps below:

1. **Data Collection:** Extraction of standardised financial statements for five FMCG companies from Screener.in into a single Excel workbook.
2. **Data Import:** Connecting Power BI Desktop to the Excel workbook using Get Data.
3. **Data Transformation:** Extraction, reshaping, and consolidation using a Python script connected as a Power BI data source, as detailed in Section 6.

4. Data Modelling: Building a star-schema model with fact and dimension tables and defining relationships.
5. DAX Measure Development: Writing calculated measures for profitability, return, leverage, coverage, valuation, and growth ratios.
6. Dashboard Design: Building KPI cards, trend visuals, comparison charts, and slicers across multiple report pages.
7. Analysis and Interpretation: Reviewing computed ratios and visual patterns to derive business insights (Section 13).
8. Validation: Cross-checking DAX-calculated ratios against manually computed values for a sample year, to ensure measure accuracy before finalising the dashboard.

This top-down workflow — source data → transformation → model → measures → visuals — mirrors standard industry practice for Power BI-based financial analytics projects and was followed strictly, with no ratio calculated outside of Power BI's DAX engine.

8. Dashboard Design (KPIs, Charts, Slicers, and Visualisations)

The Power BI report was implemented across three pages, each carrying an identical pair of slicers (Company and Financial Year) so that any selection made by the viewer stays in effect while moving between pages. Rather than a single crowded page, the ten questions from the project brief are spread across three focused pages: an overview of scale and trend, a profitability-and-valuation comparison, and a full ratio matrix alongside leverage and dividend detail.

8.1 Page 1 — Executive Overview

The top of Page 1 carries a row of six KPI cards, each bound to a single DAX measure, giving an at-a-glance summary for whichever company or companies are currently selected via the slicer:

KPI Card	DAX Measure Bound	Business Purpose
Total Sales	Total Sales	Shows revenue scale for the selected company/period
Total Market Cap	Total Market Cap	Shows market-assigned scale for the selected company
Total Net Profit	Total Net Profit	Shows absolute profitability

KPI Card	DAX Measure Bound	Business Purpose
Net Profit Margin %	Net Profit Margin %	Shows profitability efficiency, independent of company size
ROE %	ROE %	Shows return generated for equity shareholders
Debt to Equity Ratio	Debt to Equity Ratio	Shows financial leverage / balance-sheet risk

Table 8.1: KPI Cards on Page 1 — Executive Overview

Below the KPI row sit two Line Charts side by side: a Total Sales trend and a Total Net Profit trend, both plotted against Dim_Date[Date] on the X-axis with Dim_Company[Company Name] as the legend, so all five companies' trajectories can be compared on the same time axis at once.

8.2 Page 2 — Profitability, Returns and Valuation

- Clustered Column Chart, titled “Profitability vs. Shareholder Returns” — Net Profit Margin % and ROE % plotted side by side per company, deliberately pairing operating efficiency against capital efficiency so the two can be visually compared rather than read as separate numbers.
- Scatter Chart, titled “Valuation vs. Growth” — P/E Ratio on the X-axis, Sales CAGR % on the Y-axis, one bubble per company sized by Total Market Cap, giving a two-dimensional read of which companies are priced richly versus cheaply relative to how fast they are growing.

8.3 Page 3 — Leverage, Dividends and the Ratio Matrix

- Clustered Bar Chart, titled “Leverage & Debt Safety” — Debt to Equity Ratio per company, with Interest Coverage Ratio bound as a Tooltip field so hovering over any bar surfaces the debt-servicing context behind the leverage figure.
- Clustered Bar Chart, titled “Dividend Comparison” — Dividend Payout % per company.
- Matrix (pivot table) — one row per company, with ten measure columns: Total Sales, Total Net Profit, Net Profit Margin %, ROE %, Debt to Equity Ratio, Interest Coverage Ratio, EPS, P/E Ratio, Sales CAGR %, and Dividend Payout % — giving a single comprehensive reference view of every ratio discussed in Section 9, filterable by the same Company and Year slicers as the rest of the report.

8.4 Slicers and Filters

- Company Slicer (tile-style, multi-select): present on all three pages, bound to Dim_Company[Company Name]. Multi-select allows the viewer to isolate a single company or compare any subset of the five.
- Financial Year Slicer (Between-range mode): present on all three pages, bound to Dim_Date[Date], allowing the viewer to restrict every visual on the current page to a chosen year or year range — for example, narrowing to FY2026 alone to see the latest-year figures that match Section 9's tables, or widening to the full FY2017–FY2026 range to see cumulative or trend behaviour.

8.5 Interactivity Implemented

- Cross-filtering: selecting a company in the Company slicer, or clicking a data point in any chart, filters every KPI card, chart, and the Matrix table on that same page — confirmed by validating that isolating Britannia Industries Ltd. and FY2026 on the Company and Year slicers reproduces the exact 13.23% Net Profit Margin figure reported in Section 9.1.
- Consistent slicer state across pages: because the same Company and Year slicer fields are placed on every page, the analytical context a viewer sets on Page 1 carries the same intent through to Pages 2 and 3, even though each page's slicer instance is filtered independently within Power BI's default behaviour.

8.6 Planned Enhancements Not Included in the Current Build

The original design brief for this dashboard additionally specified drill-through navigation from an overview page into a single-company deep-dive page, dynamic DAX-driven page titles, a field-parameter-driven re-ranking control on the Matrix table, a custom tooltip page, and conditional-formatting colour scales on the Matrix. These were treated as a secondary phase of the build and are not present in the current .pbix file; they are carried forward as documented enhancement opportunities in Section 12.2 (Future Scope) rather than claimed as implemented features.

9. Analysis and Findings

This section presents the actual computed ratios from the dataset, as reflected on the Power BI dashboard, for the most recently reported financial year of each company (FY2025–26 for

Britannia, HUL, ITC and Nestlé; CY2025 for Varun Beverages, which follows a December year-end).

9.1 Profitability Ratios

Company	Sales (₹ Cr)	Net Profit (₹ Cr)	Net Profit Margin %	Operating (EBITDA) Margin %
Britannia	19,151.6	2,533.5	13.23%	18.12%
HUL	64,468.0	15,040.0	23.33%	22.00%
ITC	78,868.4	20,689.5	26.23%	28.15%
Nestlé India	23,154.6	3,544.6	15.31%	21.47%
Varun Beverages	21,685.4	3,036.5	14.00%	23.46%

Table 9.1: Latest-Year Profitability Ratios (computed via DAX)

ITC records the highest Net Profit Margin (26.23%) and Operating Margin (28.15%) among the five companies, reflecting the high-margin tobacco business embedded within its diversified FMCG portfolio. HUL's Net Profit Margin of 23.33% is markedly higher than its typical trend of around 16–17% seen from FY2021–FY2025, on account of a one-time other-income item of ₹4,923 Cr recorded in FY2026. Britannia posts the lowest Net Profit Margin (13.23%) among the five, consistent with the thinner margins typical of a pure biscuits-and-bakery business exposed to raw-material (edible oil, wheat, sugar) cost volatility.

9.2 Return Ratios

Company	Return on Equity (ROE) %	Return on Capital Employed (ROCE) %
Britannia	49.61%	52.44%
HUL	30.86%	37.10%
ITC	28.53%	37.54%
Nestlé India	66.76%	82.87%
Varun Beverages	15.51%	19.04%

Table 9.2: Latest-Year Return Ratios (computed via DAX)

Nestlé India shows the strongest capital efficiency, with ROE of 66.76% and ROCE of 82.87%, a reflection of its asset-light, high-brand-equity business model requiring relatively low reinvested capital per rupee of profit. Britannia also delivers strong returns (ROE 49.61%, ROCE 52.44%), while Varun Beverages, being a capital-intensive bottling and distribution business with continuous capacity expansion, shows comparatively lower return ratios (ROE 15.51%, ROCE

19.04%) despite its rapid growth, since large recent capital expenditure has expanded its equity/asset base faster than near-term profit.

9.3 Leverage and Coverage Ratios

Company	Debt-to-Equity Ratio	Interest Coverage Ratio (x)
Britannia	0.27	30.17
HUL	0.03	45.44
ITC	0.03	330.15
Nestlé India	0.08	30.12
Varun Beverages	0.13	21.46

Table 9.3: Latest-Year Leverage and Coverage Ratios (computed via DAX)

All five companies operate with very low financial leverage, a hallmark of the FMCG sector's low capital-intensity and strong internal cash generation. ITC's Debt-to-Equity of 0.03 and an exceptionally high Interest Coverage Ratio of 330x reflect an almost debt-free balance sheet. Britannia carries the highest relative leverage (D/E of 0.27) among the five, largely used to fund capacity expansion, though its Interest Coverage Ratio of over 30x indicates negligible debt-servicing risk.

9.4 Valuation and Growth Ratios

Company	EPS (₹)	Market Price (₹)	P/E Ratio (x)	Sales CAGR %	Net Profit CAGR %
Britannia	105.17	5,423.00	51.57	8.68%	12.40%
HUL	64.01	2,055.20	32.11	7.67%	14.42%
ITC	16.51	287.70	17.42	7.04%	8.07%
Nestlé India	18.38	1,174.80	63.91	9.77%	12.53%
Varun Beverages	8.98	489.85	54.56	21.14%	60.74%

Table 9.4: EPS, P/E, and CAGR over the Study Period (computed via DAX)

Varun Beverages stands out with an exceptional Sales CAGR of 21.14% and Net Profit CAGR of 60.74% over the study period — by far the fastest-growing company in the group — reflecting rapid geographic expansion of PepsiCo's bottling franchise across India and international territories. Nestlé India and Britannia trade at the richest valuations (P/E of 63.91x and 51.57x respectively), consistent with the market's willingness to pay a premium for their consistently high

return ratios. ITC trades at the lowest P/E (17.42x) despite strong profitability, a valuation gap commonly attributed to its diversified, partly tobacco-linked business mix.

9.5 Dividend Payout

Company	Dividend Payout Ratio %
Britannia	86.05%
HUL	64.06%
ITC	87.81%
Nestlé India	65.28%
Varun Beverages	16.71%

Table 9.5: Latest-Year Dividend Payout Ratio (computed via DAX)

ITC (87.81%) and Britannia (86.05%) return the largest share of profits to shareholders as dividends, consistent with their status as mature, cash-generative businesses with limited incremental capital expenditure needs. Varun Beverages retains the vast majority of its profit (payout of only 16.71%), reinvesting internally accrued cash into its ongoing capacity-expansion programme rather than distributing it — consistent with its position as the fastest-growing company in the peer set.

10. Business Insights and Recommendations

10.1 Key Business Insights

- The FMCG peer set operates with structurally low leverage (D/E consistently below 0.3x), confirming the sector's reliance on internal accruals rather than borrowed capital for growth.
- Nestlé India and Britannia combine high margins with high capital efficiency, making them the strongest performers on a pure return-on-capital basis, and this is reflected in the dashboard's premium P/E multiples for both.
- Varun Beverages is the clear growth outlier, with Sales and Net Profit CAGR far exceeding peers; its lower ROE/ROCE and dividend payout reflect a business still in an active capital-deployment phase rather than a weaker underlying franchise.
- Both HUL (FY2026) and ITC (FY2025) show a sharp, one-year spike in Net Profit Margin driven by unusually large “Other Income” entries, most plausibly linked to one-off corporate-restructuring or asset-monetisation gains rather than core operating performance — a pattern

the dashboard's year-on-year trend line makes easy to spot and which should be adjusted for when comparing profitability across years.

- ITC's near-zero leverage and 330x interest coverage indicate significant unused balance-sheet capacity, which the company could deploy for acquisitions or accelerated FMCG (ex-cigarettes) expansion without meaningfully raising financial risk.

10.2 Recommendations

- For investors seeking capital-efficient compounding, Nestlé India and Britannia merit closer attention, given their consistently superior ROE/ROCE profile across the ten-year study period, subject to their relatively higher valuations.
- For investors seeking growth exposure within FMCG, Varun Beverages represents the strongest growth trajectory in the group; its comparatively modest dividend payout should be read as reinvestment for future growth rather than a negative signal.
- When comparing year-on-year profitability across companies, one-off items (as seen in HUL FY2026 and ITC FY2025) should be normalised for, to avoid over- or under-stating a company's sustainable earnings power — a recommended enhancement is a dedicated “Adjusted Net Profit” DAX measure that flags/excludes known exceptional items.
- Given its very low leverage, ITC has meaningful headroom to pursue debt-funded expansion of its non-cigarette FMCG portfolio, which could support long-term multiple re-rating.

11. Limitations and Future Scope

11.1 Limitations

- The dataset relies on a single third-party aggregator (Screener.in); minor reclassification differences versus companies' own audited annual reports cannot be ruled out.
- Current Ratio and Quick Ratio could not be computed with full precision, as the source balance sheet groups current and non-current liabilities together under a single “Other Liabilities” line rather than separating them; this is disclosed as a scope limitation rather than approximated with assumed figures.

- Nestlé India's transition from a December to a March financial-year-end (visible as a stub reporting period in FY2023–24) and Varun Beverages' continued December year-end (versus the March year-end of the other four companies) introduce minor period-alignment differences when comparing “latest year” figures directly.
- One-off items embedded within “Other Income” (e.g., the FY2026 HUL and FY2025 ITC spikes) were identified through pattern analysis of the trend visuals but could not be independently confirmed against the underlying corporate-action disclosures, since only Screener-exported P&L/Balance Sheet/Cash Flow data was available for this project.
- The dashboard reflects data as of the most recent extraction date; it is not connected to a live/real-time data feed.
- Data loading depends on a local Python installation (with pandas and openpyxl) being available to Power BI Desktop; on a machine without Python configured, the workbook must instead be loaded through the manual Power Query steps as a fallback.

11.2 Future Scope

- Implement drill-through navigation from Page 1 into a dedicated single-company deep-dive page, along with a DAX-driven dynamic page title — both were part of the original dashboard design but were not completed in the current .pbix build.
- Add a field-parameter-driven “Rank By” control and colour-scale conditional formatting to the Matrix table on Page 3, so a viewer can re-sort the same table by any ratio and instantly see the strongest/weakest performer highlighted.
- Add a custom tooltip report page (e.g., a mini Net Profit Margin % trend line) bound to the Page 2 comparison charts, to surface additional context on hover without adding further visuals to the canvas.
- Extend the fact-table model to include additional FMCG peers (e.g., Dabur, Marico, Colgate-Palmolive India) for a broader industry benchmark.
- Integrate a live data-refresh pipeline using Power BI Service and a scheduled connector to Screener.in or an equivalent financial data API.
- Add a dedicated “Adjusted Profitability” DAX measure set that flags and normalises for known exceptional/one-off items across years.

- Incorporate sector-level and Nifty FMCG Index benchmarking to contextualise each company's ratios against the broader market.
- Add predictive elements (e.g., Power BI's built-in forecasting or an integrated Python/R visual) to project near-term sales and profit trends.

12. Conclusion

This project successfully demonstrates the application of Power BI — spanning a Python-scripted data pipeline, star-schema data modelling, DAX-based ratio computation, and interactive dashboard design — to convert ten years of raw financial statement data for five leading Indian FMCG companies into a decision-useful business intelligence tool.

The analysis confirms that the Indian FMCG sector, as represented by Britannia, HUL, ITC, Nestlé India, and Varun Beverages, is characterised by strong profitability, low financial leverage, and healthy interest coverage, while also showing meaningful divergence in growth rates and capital efficiency across companies — most notably the exceptional growth profile of Varun Beverages against the capital-efficient, higher-margin profiles of Nestlé India and Britannia.

Beyond the financial findings, the project provided practical, end-to-end experience in building a professional-grade Power BI solution: programmatically reshaping multi-block source data, designing a scalable star-schema data model, writing and validating DAX measures, and creating a three-page interactive dashboard with slicers and cross-filtering — skills directly relevant to a career at the intersection of data, technology, and finance.

13. References

1. Screener.in — Financial statement data (Profit & Loss, Balance Sheet, Cash Flow) for Britannia Industries Ltd., Hindustan Unilever Ltd., ITC Limited, Nestlé India Ltd., and Varun Beverages Ltd.
2. Microsoft Power BI Documentation — Power Query, Data Modelling, and Microsoft Learn.
3. Annual Reports and Investor Presentations of the respective companies (as consolidated on Screener.in).
4. Course materials — “Data Analysis and Visualization with MS Excel and Power BI,” Lovely Professional University.

14. Appendix

A. Dashboard Page Layouts

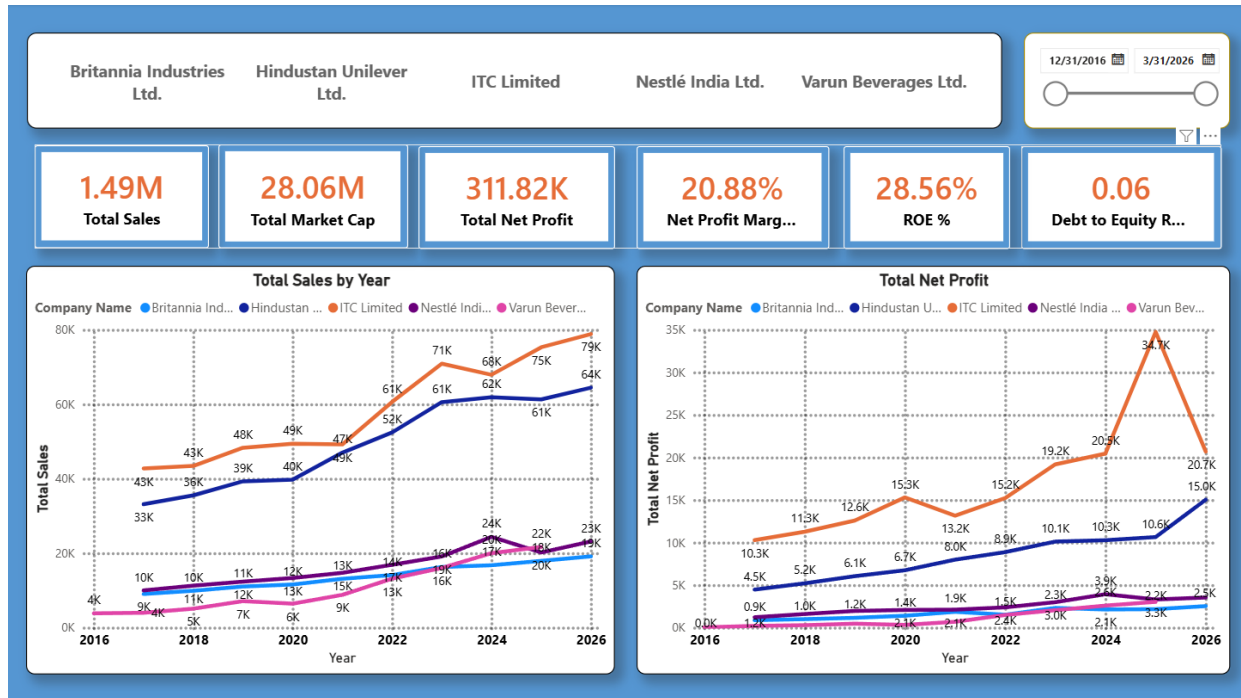


Figure 14.1: Page 1 — Executive Overview (KPI row, Company/Year slicers, Sales and Net Profit trend charts)



Figure 14.2: Page 2 — Profitability vs. Shareholder Returns and Valuation vs. Growth

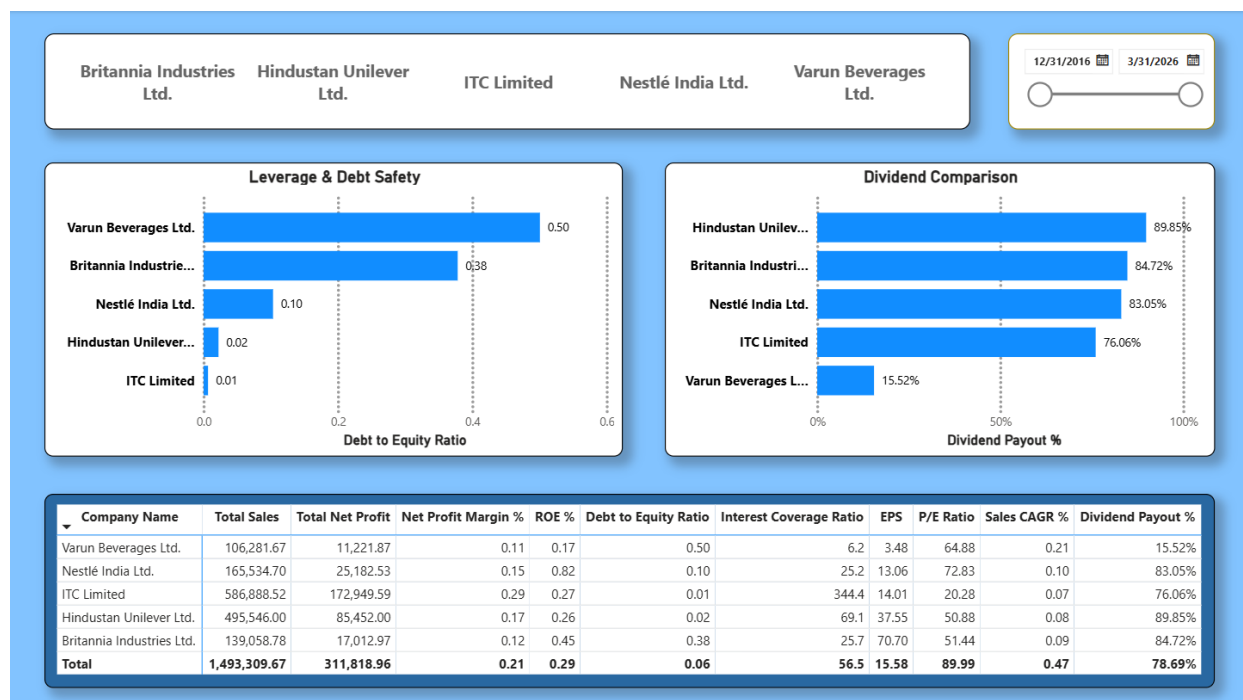


Figure 14.3: Page 3 — Leverage & Debt Safety, Dividend Comparison, and the full Ratio Matrix

B. Key DAX Measures Implemented

Measure Name	DAX Formula (as implemented)	Purpose
Total Sales	Total Sales = SUM(Fact_ProfitLoss[Sales])	Aggregates revenue for selected company/period
Total Net Profit	Total Net Profit = SUM(Fact_ProfitLoss[Net Profit])	Aggregates bottom-line profit
Net Profit Margin %	NPM % = DIVIDE([Total Net Profit],[Total Sales],0)	Profitability relative to revenue
Total Equity	Total Equity = SUM(Fact_BalanceSheet[Equity Share Capital]) + SUM(Fact_BalanceSheet[Reserves])	Shareholders' fund base for return ratios (helper measure)
ROE %	ROE % = DIVIDE([Total Net Profit],[Total Equity],0)	Return generated on shareholders' equity
Total Borrowings	Total Borrowings = SUM(Fact_BalanceSheet[Borrowings])	Aggregated debt figure (helper measure)
Debt to Equity Ratio	D/E = DIVIDE([Total Borrowings],[Total Equity],0)	Financial leverage / risk

Measure Name	DAX Formula (as implemented)	Purpose
Interest Coverage Ratio	$\text{ICR} = \text{DIVIDE}(\text{SUM}(\text{Fact_ProfitLoss}[\text{Profit before tax}]) + \text{SUM}(\text{Fact_ProfitLoss}[\text{Interest}]), \text{SUM}(\text{Fact_ProfitLoss}[\text{Interest}]), 0)$	Comfort in servicing interest obligations
Market Price	$\text{Market Price} = \text{AVERAGE}(\text{Fact_BalanceSheet}[\text{Market Price}])$	Latest traded price (helper measure)
Adjusted Shares (Cr)	$\text{Adjusted Shares (Cr)} = \text{SUM}(\text{Fact_BalanceSheet}[\text{Adjusted Equity Shares in Cr}])$	Share count base for per-share measures (helper measure)
EPS	$\text{EPS} = \text{DIVIDE}([\text{Total Net Profit}], [\text{Adjusted Shares (Cr)}], 0)$	Per-share earnings for valuation comparison
P/E Ratio	$\text{P/E} = \text{DIVIDE}([\text{Market Price}], [\text{EPS}], 0)$	Market valuation relative to earnings
Total Market Cap	$\text{Total Market Cap} = [\text{Market Price}] * [\text{Adjusted Shares (Cr)}]$	Market-assigned company scale
Dividend Payout %	$\text{Payout \%} = \text{DIVIDE}(\text{SUM}(\text{Fact_ProfitLoss}[\text{Dividend Amount}]), [\text{Total Net Profit}], 0)$	Share of profit distributed to shareholders
Sales CAGR %	Context-safe CAGR using each company's own first/last reported year in Fact_ProfitLoss (see formula note below)	Compounded annual growth rate of revenue, correctly scoped per company

Table 14.1: DAX Measures Implemented in the Power BI Model

Note on Sales CAGR %: an initial version of this measure used `FIRSTDATE(Dim_Date[Date]) / LASTDATE(Dim_Date[Date])`, which incorrectly anchored every company to the calendar's overall earliest and latest dates (2016 and 2026) rather than to each company's own first and last reported year — producing a distorted result for companies such as Britannia, HUL, ITC, and Nestlé India, which report from FY2017 onward rather than 2016. The corrected measure instead derives each company's own first and last Report Date directly from Fact_ProfitLoss within the current filter context, so the scatter chart on Page 2 shows each company's true, individually-scoped growth rate rather than one distorted by Varun Beverages' earlier 2016 start date.

The measures ROCE %, Operating Margin % (EBITDA-based), and time-intelligence growth measures (Sales YoY Growth %, Net Profit CAGR %) discussed in the original design documentation for this project were not included in the current dashboard build; they remain candidates for the future enhancements listed in Section 12.2.